Robed Beauvile-Owen, Ph.D.

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses, turning owners' growth goals into site-selection analyses and revenue targets; identified a $1.5M expansion opportunity.
- Built a revenue driver tree (traffic × capture rate × ticket price, with a traffic-to-capture elasticity) and a margin model (fixed and variable costs per car) that translated client profit goals into break-even revenue and volume targets.
- Estimated the incremental revenue of new locations with Double ML and causal forests (econml, XGBoost nuisance models) on a 12-year county panel, stating identification assumptions and checking robustness with difference-in-differences and a synthetic-data recovery test.
- Back-tested demand predictions out of sample against real operating sites (bias, error, rank correlation, 90% interval coverage).
- Built low/base/high scenarios and Monte Carlo risk models (probability of loss, VaR, CVaR) that carry the causal estimates' 95% confidence intervals, and explained sensitivity and uncertainty to owners in written reports.
- Engineered reproducible Python pipelines with data-quality checks and 150+ unit tests, cutting analysis turnaround by 91%, and documented each method in reusable walkthroughs.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap with strategy, finance, engineering, and behavioral science leadership, owning the acquisition KPI framework: gross lead rate (GLR), member rate, and cost per member (CAC).
- Built a causal graph (DAG) from historical campaign data showing GLR drives member rate, which drives cost per member, focusing model building and causal tests on GLR as the lever.
- Built an in-house experimentation tool and standardized A/B testing methods that data science and strategy used to validate lift across prospect segments and set each campaign's ad-spend budget, with CUPED cutting variance 20%.
- Built uplift models on millions of HIPAA-protected (PII/PHI) Medicare/Medicaid-eligible records to target incremental (causal) response rather than predicted response, identifying one client's "Opportunist" segment as the main driver of GLR and cutting ad spend by $10K per campaign.
- Linked the monthly retention survey and member funnel to retention behavior, finding claims and ticket complaints were the main drivers of churn.
- Owned the monthly model monitoring report on drift, performance, and media mix forecast-vs-actual accuracy for up to 10 production models for the directors of data science and strategy and the managing partner, plus a company-wide self-service dashboard refreshed by Databricks jobs.
- Re-architected a production churn prediction platform in Python, SQL, Spark, Snowflake, and Databricks, reducing runtime by 48% and memory use by 23%.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics, evaluation frameworks, and drift monitoring for production search and pricing models, using offline validation and online experiments, and mentored the team on ML practices.
- Built SQL training tables that broke bulk orders into item-level shipping costs, then partnered with engineering to ship a regression-based pricing API in C# on Azure, scaling to 10,000+ daily requests.
- Reduced customer dissatisfaction with search by 12% with product and engineering partners through custom Elasticsearch indexing, tokenizers, and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built and validated XGBoost classification models on millions of consumer records with hundreds of external attributes, driving a 3% lift in response rates, and explained model drivers to stakeholders with SHAP.
- Automated reporting pipelines and Power BI dashboards with Python, SQL, and Snowflake, removing 60 minutes of manual work daily, and improved geospatial targeting accuracy by 20% with GeoPandas.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in optimization, functional analysis, and operator theory, and taught mathematics and statistics to classes of 75-200 students.

TECHNICAL PROJECTS

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes to show when the LTV = ARPU / churn identity breaks down, estimating LTV from survival curves instead.

Mobile Game Retention A/B Test github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Ran a full experiment workflow on 1- and 7-day player retention: hypotheses, outlier handling, sample sizing, significance testing, and practical significance for the launch call.

Medicare Claims Intelligence & Fraud Detection github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a tested dbt warehouse (staging to marts) over 790K CMS claims and a provider fraud detector with 81% precision, cut forecast error 41% vs. a seasonal benchmark in rolling backtests (ARIMA, ETS, Prophet), and built a SHAP-explained model that finds 46% more high-cost members.

Starbucks Rewards: Predicting Offer Responses github.com/beauvilerobed/predict-consumer-responses

- Predicted how rewards app members respond to BOGO, discount, and informational offers from transaction, offer, and demographic data, with an XGBoost multi-class model trained and deployed on AWS SageMaker.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Metrics & Planning: KPI Trees & Driver Decomposition, Metric Definitions, Acquisition Funnel & CAC, Retention & Cohort Analysis, Survival Analysis, LTV / ARPU, Sensitivity & Scenario Analysis, Target Setting, Forecast Validation (Backtesting)

Causal Inference & Experimentation: A/B Testing, CUPED, Power Analysis, Double ML, Causal Forests, Difference-in-Differences, CATE / Uplift Modeling, Robustness Checks, Monte Carlo Simulation

Modeling: Regression, Classification, XGBoost, Time Series Forecasting (ARIMA, ETS, Prophet), Media Mix Modeling, Causal Graphs (DAGs), Graph Mining (NetworkX), SHAP, Model Monitoring & Drift

Data & Tools: SQL, Python (pandas, NumPy, statsmodels, scikit-learn, econml), dbt, Snowflake, Spark, Databricks, SQL Server, PostgreSQL, AWS, Azure, Git, Unit Testing, Power BI, Tableau, Excel, AI-Assisted Development (Claude Code)